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Studierichting: Informatica
Oefeningenreeks 4A nr. 42.8

42.8 $\displaystyle\int \frac{dx}{x^3 - 7x + 6}$

$x^3 - 7x + 6 = (x-1)(x-2)(x+3)$

$$\frac{1}{x^3 - 7x + 6} = \frac{1}{(x-1)(x-2)(x+3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x+3}$$

$$A = \left.\frac{1}{(x-2)(x+3)}\right|_{x=1} = \frac{1}{(-1)(4)} = -\frac{1}{4}$$

$$B = \left.\frac{1}{(x-1)(x+3)}\right|_{x=2} = \frac{1}{(1)(5)} = \frac{1}{5}$$

$$C = \left.\frac{1}{(x-1)(x-2)}\right|_{x=-3} = \frac{1}{(-4)(-5)} = \frac{1}{20}$$

$$\frac{1}{x^3 - 7x + 6} = -\frac{1}{4(x-1)} + \frac{1}{5(x-2)} + \frac{1}{20(x+3)}$$

$$\int \frac{dx}{x^3 - 7x + 6} = -\frac{1}{4}\ln|x-1| + \frac{1}{5}\ln|x-2| + \frac{1}{20}\ln|x+3| + c$$

References